Unit 3: Data Link Layer

In the data link layer of the OSI model, framing refers to the process of dividing a stream of data bits into manageable units called frames. Frames are used to encapsulate the data being transmitted over the physical medium and include additional control information for error detection, flow control, and synchronization.

Framing is necessary because data transmission in the data link layer typically occurs in the form of a continuous bit stream. Without framing, the receiving device wouldn't know where one frame ends and the next one begins. By adding a header and a trailer to the data, framing creates a recognizable structure that enables the receiver to identify individual frames.

The framing process involves the following steps:

1. Character/byte stuffing: In some framing techniques, special control characters are used to denote the start and end of a frame. To prevent these control characters from being mistaken as actual data, byte stuffing is performed. It involves adding an extra byte or escaping specific control characters within the data stream.

2. Bit stuffing: In other framing techniques, bit stuffing is used to delimit frames. A predefined pattern of bits is inserted into the data stream whenever a specific bit sequence appears. This ensures that the receiver can identify the boundaries of the frames by recognizing the stuffing pattern.

3. Frame header: Each frame begins with a header that contains control information such as source and destination addresses, frame sequence numbers, and error detection bits. This header helps the receiving device interpret the frame and perform necessary actions.

4. Frame trailer: The frame ends with a trailer, which typically includes error detection and error correction information, such as a cyclic redundancy check (CRC) or checksum. The receiver can use this information to verify the integrity of the received frame and request retransmission if errors are detected.

By using framing, the data link layer ensures reliable and orderly transmission of data between network nodes.

- **flow control and error control protocols in data link layer**

In the data link layer of the OSI model, flow control and error control protocols are used to ensure reliable and efficient transmission of data between two adjacent nodes, typically over a physical link or channel. Let's discuss each of these protocols separately:

1. Flow Control Protocols: Flow control protocols are responsible for managing the flow of data between the sender and receiver to avoid overwhelming the receiver with more data than it can handle. They help prevent data loss or degradation due to congestion or buffer overflow. Two commonly used flow control protocols are:

   a. Stop-and-Wait: In this protocol, the sender sends one frame at a time and waits for an acknowledgment (ACK) from the receiver before sending the next frame. This ensures that the receiver can handle and process each frame before the sender sends the next one. It provides simple and reliable flow control but can be inefficient for long delays in the round-trip time.
   b. Sliding Window: The sliding window protocol allows the sender to transmit multiple frames without waiting for individual acknowledgments. The sender maintains a sliding window of acceptable sequence numbers, indicating the range of frames that can be sent. The receiver sends acknowledgment messages for received frames, and the sender adjusts the window based on these acknowledgments. It enables better link utilization and improves efficiency compared to stop-and-wait.
2. Error Control Protocols: Error control protocols are used to detect and correct errors that may occur during data transmission, ensuring data integrity. These protocols typically employ techniques such as error detection codes, retransmission, and acknowledgment mechanisms. Two common error control protocols are:

a. Automatic Repeat reQuest (ARQ): ARQ protocols are used to ensure reliable delivery of data by requesting retransmission of lost or corrupted frames. There are several types of ARQ protocols, including:

- Stop-and-Wait ARQ: Similar to the flow control protocol, the sender waits for an acknowledgment before sending the next frame. If the sender does not receive an acknowledgment within a specified timeout period, it retransmits the frame.
- Go-Back-N ARQ: The sender can transmit multiple frames without waiting for individual acknowledgments. The receiver acknowledges the receipt of a frame and maintains a window of acceptable sequence numbers. If a frame is lost or corrupted, the receiver discards it and asks the sender to retransmit all the frames in the window from the lost/corrupted frame onwards.
- Selective Repeat ARQ: Similar to Go-Back-N, the sender can transmit multiple frames, and the receiver maintains a window. However, in Selective Repeat, the receiver acknowledges the receipt of each frame individually, allowing the sender to retransmit only the lost or corrupted frames.

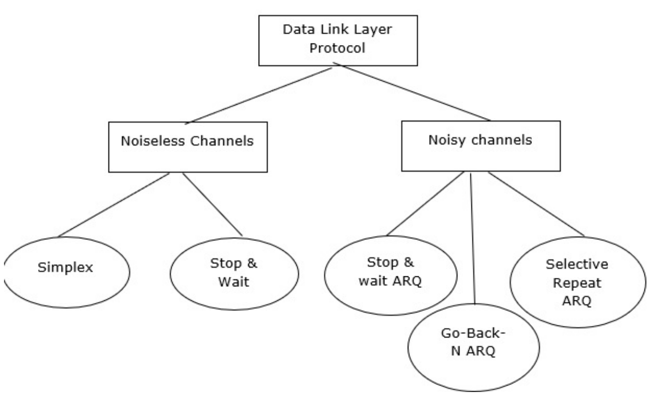
b. Error Detection Codes: Error detection codes, such as cyclic redundancy check (CRC) or checksum, are added to the data frames before transmission. The receiver calculates the error detection code from the received data and compares it with the received code. If they don't match, an error is detected, and the receiver can request retransmission of the frame.

These protocols work together to ensure reliable and efficient data transmission in the data link layer. Flow control manages the pace of data transmission, while error control detects and corrects transmission errors to maintain data integrity.

- **<u>Noisy and Noiseless channels</u>**

Data link layer protocols are divided into two categories based on whether the transmission channel is noiseless or noisy.

The data link layer protocol is diagrammatically represented below −



Noiseless Channels

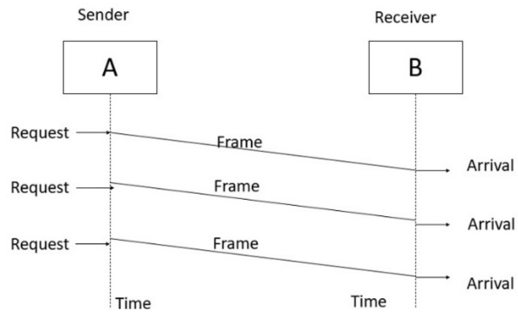
There are two noiseless channels which are as follows −

Simplest Protocol

Step 1 − Simplest protocol that does not have flow or error control.

Step 2 − It is a unidirectional protocol where data frames are traveling in one direction that is from the sender to receiver.

Step 3 − Let us assume that the receiver can handle any frame it receives with a processing time that is small enough to be negligible, the data link layer of the receiver immediately removes the header from the frame and hands the data packet to its network layer, which can also accept the packet immediately.



Stop-and-Wait Protocol

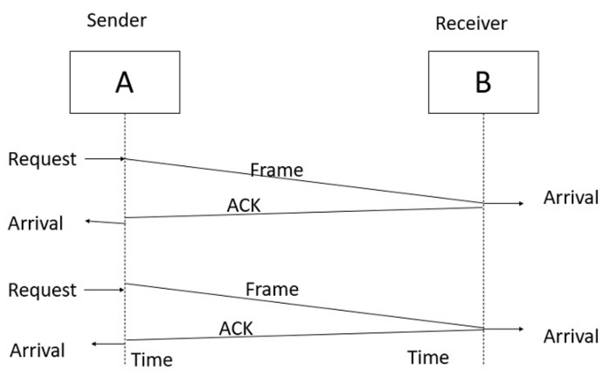
Step 1 − If the data frames that arrive at the receiver side are faster than they can be processed, the frames must be stored until their use.

Step 2 − Generally, the receiver does not have enough storage space, especially if it is receiving data from many sources. This may result in either discarding of frames or denial of service.

Step 3 − To prevent the receiver from becoming overwhelmed with frames, the sender must slow down. There must be ACK from the receiver to the sender.

Step 4 − In this protocol the sender sends one frame, stops until it receives confirmation from the receiver, and then sends the next frame.

Step 5 − We still have unidirectional communication for data frames, but auxiliary ACK frames travel from the other direction. We add flow control to the previous protocol.



Noisy Channels

There are three types of requests for the noisy channels, which are as follows −

- Stop & wait Automatic Repeat Request.
- Go-Back-N Automatic Repeat Request.
- Selective Repeat Automatic Repeat Request.

Noiseless channels are generally non-existent channels. We can ignore the error or we need to add error control to our protocols.

Stop and Wait Automatic Repeat Request

Step 1 − In a noisy channel, if a frame is damaged during transmission, the receiver will detect with the help of the checksum.

Step 2 − If a damaged frame is received, it will be discarded, and the transmitter will retransmit the same frame after receiving a proper acknowledgement.

Step 3 − If the acknowledgement frame gets lost and the data link layer on 'A' eventually times out. Not having received an ACK, it assumes that its data frame was lost or damaged and sends the frame containing packet 1 again. This duplicate frame also arrives at the data link layer on 'B', thus part of the file will be duplicated and protocol is said to be failed.

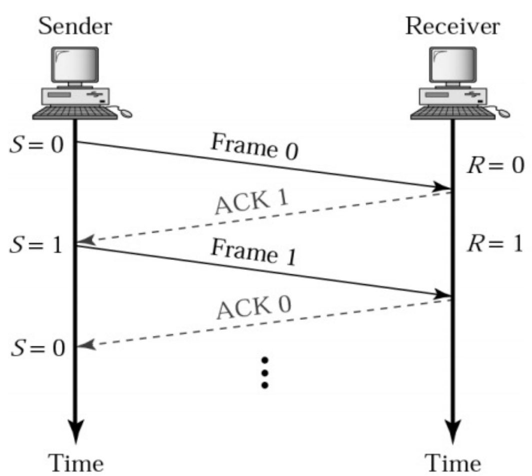
Step 4 − To solve this problem, assign a sequence number in the header of the message.

Step 5 − The receiver checks the sequence number to determine if the message is a duplicate since only the message is transmitted at any time.

Step 6 − The sending and receiving station needs only a 1-bit alternating sequence of '0' or '1' to maintain the relationship of the transmitted message and its ACK/ NAK.

Step 7 − A modulo-2 numbering scheme is used where the frames are alternatively labelled with '0' or '1' and positive acknowledgements are of the form ACK 0 and ACK 1.

Normal operation of Stop & Wait ARQ is given below −



Go-Back-N ARQ

To improve the transmission efficiency, we need more than one frame to be outstanding to keep the channel busy while the sender is waiting for acknowledgement.

There are two protocols developed for achieving this goal and they are as follows −

- Go – Back - N – Automatic – Repeat Request
- Sliding window protocol

Go-Back-N ARQ

Step 1 − In this protocol we can send several frames before receiving acknowledgements.

Step 2 − we keep a copy of these frames until the acknowledgment arrives.

Step 3 − Frames from a sending station are numbered sequentially. However, we need to include the sequence number of each frame in the header; we need to set a limit.

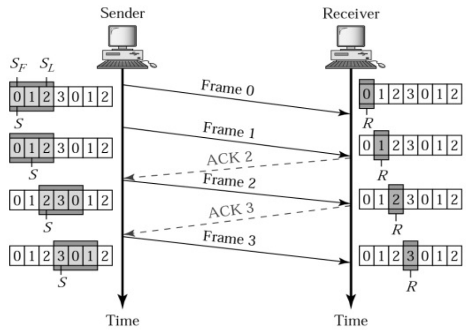
Step 4 − If the header of the frame allows m bits for the sequence number, the sequence numbers range from 0 to 2^m-1. We can also repeat the sequence numbers.

Example

For m = 2, the range of sequence numbers is: 0 to 3, i.e.

0,1,2,3, 0,1,2,3,…

The Go-Back-N ARQ is shown below in diagram format −



- **HDLC**

HDLC (High-Level Data Link Control) is a protocol used in computer networks for reliable and efficient data communication between network nodes. It operates at the data link layer of the OSI model and provides a framework for framing, error detection, and flow control.

HDLC Modes:

1. Normal Response Mode (NRM): In NRM, one device acts as the primary station (typically the sender) while the other device functions as the secondary station (typically the receiver). The primary station initiates frame transmission, and the secondary station responds with acknowledgments or requests for retransmission. NRM allows for full-duplex communication.

2. Asynchronous Response Mode (ARM): ARM is a half-duplex mode of HDLC where both stations can initiate frame transmission. Each frame includes an address field, allowing the receiving station to identify the intended recipient. ARM does not support simultaneous bidirectional data transfer.

3. Asynchronous Balanced Mode (ABM): ABM is an extended version of ARM that supports bidirectional communication between two equal devices (no primary or secondary designation). ABM introduces a polling mechanism to avoid collisions. Each station takes turns acting as a master and polls the other station for data transmission.

HDLC Frame Structure:

An HDLC frame consists of the following components:

1. Flag: The frame starts and ends with a flag sequence (01111110), which serves as a delimiter to indicate the beginning and end of the frame.

2. Address: The address field identifies the destination or source station, depending on the mode of operation. In NRM, it is typically the secondary station's address.

3. Control: The control field carries information about the frame type, sequence numbers, and other control parameters, depending on the specific mode and options used.

4. Data: The data field contains the payload or actual data being transmitted. Its length can vary, depending on the data being sent.

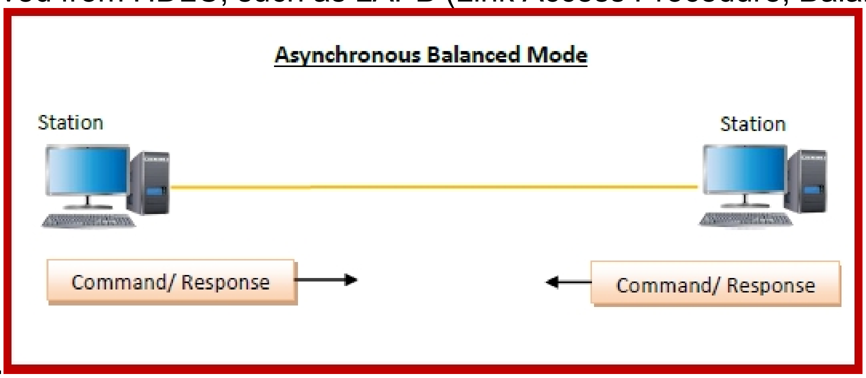
5. Frame Check Sequence (FCS): The FCS field is used for error detection. It contains a cyclic redundancy check (CRC) value computed over the entire frame, allowing the receiver to verify the integrity of the received data.

6. Optional Fields: HDLC allows for additional optional fields like information field extension, quality indicator, and padding to align frames with byte boundaries.

HDLC frames are transmitted bit by bit, with bit stuffing employed to ensure frame synchronization and distinguish between data and control characters. Bit stuffing involves inserting an extra 0 bit after every consecutive sequence of five 1 bits in the data stream to avoid confusion with the flag sequence.

The receiver uses the FCS field to perform error detection. If the calculated CRC does not match the received FCS value, an error is detected, and the frame is discarded or flagged for retransmission.

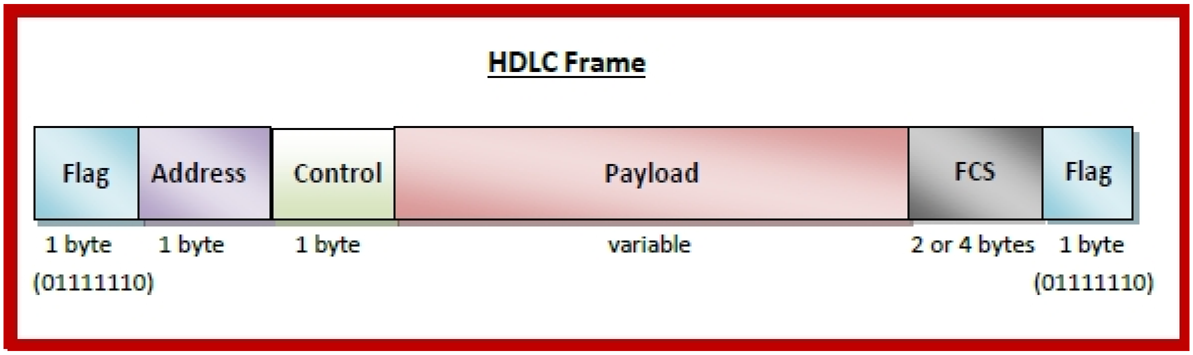
HDLC is widely used in various network technologies, including point-to-point links, WAN connections, and protocols derived from HDLC, such as LAPB (Link Access Procedure, Balanced) and PPP (Point-to-Point Protocol).



HDLC Frame

HDLC is a bit - oriented protocol where each frame contains up to six fields. The structure varies according to the type of frame. The fields of a HDLC frame are −

- **Flag** − It is an 8-bit sequence that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.
- **Address** − It contains the address of the receiver. If the frame is sent by the primary station, it contains the address(es) of the secondary station(s). If it is sent by the secondary station, it contains the address of the primary station. The address field may be from 1 byte to several bytes.
- **Control** − It is 1 or 2 bytes containing flow and error control information.
- **Payload** − This carries the data from the network layer. Its length may vary from one network to another.
- **FCS** − It is a 2 byte or 4 bytes frame check sequence for error detection. The standard code used is CRC (cyclic redundancy code)



- **PPP**

Point - to - Point Protocol (PPP) is a communication protocol of the data link layer that is used to transmit multiprotocol data between two directly connected (point-to-point) computers. It is a byte - oriented protocol that is widely used in broadband communications having heavy loads and high speeds. Since it is a data link layer protocol, data is transmitted in frames. It is also known as RFC 1661.

Services Provided by PPP

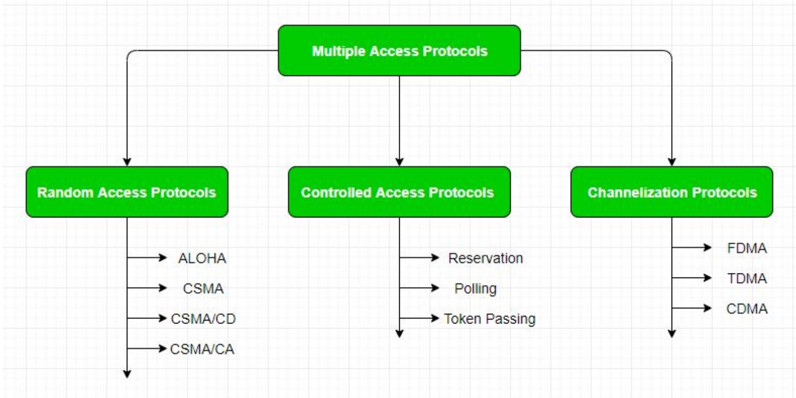
The main services provided by Point - to - Point Protocol are −

- Defining the frame format of the data to be transmitted.
- Defining the procedure of establishing link between two points and exchange of data.
- Stating the method of encapsulation of network layer data in the frame.
- Stating authentication rules of the communicating devices.
- Providing address for network communication.
- Providing connections over multiple links.
- Supporting a variety of network layer protocols by providing a range os services.

- **Multiple Access Control**

Multiple Access Control –
If there is a dedicated link between the sender and the receiver then data link control layer is sufficient, however if there is no dedicated link present then multiple stations can access the channel simultaneously. Hence multiple access protocols are required to decrease collision and avoid crosstalk. For example, in a classroom full of students, when a teacher asks a question and all the students (or stations) start answering simultaneously (send data at same time) then a lot of chaos is created(data overlap or data lost) then it is the job of the teacher (multiple access protocols) to manage the students and make them answer one at a time.

Multiple access protocols can be subdivided further as –



1. Random Access Protocol: In this, all stations have same superiority that is no station has more priority than another station. Any station can send data depending on medium's state(idle or busy). It has two features:

There is no fixed time for sending data

There is no fixed sequence of stations sending data

The Random-access protocols are further subdivided as:

(a) ALOHA – It was designed for wireless LAN but is also applicable for shared medium. In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.

Pure Aloha:
When a station sends data it waits for an acknowledgement. If the acknowledgement doesn't come within the allotted time, then the station waits for a random amount of time called back-off time (Tb) and re-sends the data. Since different stations wait for different amount of time, the probability of further collision decreases.

Vulnerable Time = 2* Frame transmission time

Throughput = G exp{-2*G}

Maximum throughput = 0.184 for G=0.5

Slotted Aloha:
It is similar to pure aloha, except that we divide time into slots and sending of data is allowed only at the beginning of these slots. If a station misses out the allowed time, it must wait for the next slot. This reduces the probability of collision.

Vulnerable Time = Frame transmission time

Throughput = G exp{-*G}

Maximum throughput = 0.368 for G=1

(b) CSMA – Carrier Sense Multiple Access ensures fewer collisions as the station is required to first sense the medium (for idle or busy) before transmitting data. If it is idle then it sends data, otherwise it waits till the channel becomes idle. However there is still chance of collision in CSMA due to propagation delay. For example, if station A wants to send data, it will first sense the medium.If it finds the channel idle, it will start sending data. However, by the time the first bit of data is transmitted (delayed due to propagation delay) from station A, if station B requests to send data and senses the medium it will also find it idle and will also send data. This will result in collision of data from station A and B.

CSMA access modes-

1-persistent: The node senses the channel, if idle it sends the data, otherwise it continuously keeps on checking the medium for being idle and transmits unconditionally(with 1 probability) as soon as the channel gets idle.

Non-Persistent: The node senses the channel, if idle it sends the data, otherwise it checks the medium after a random amount of time (not continuously) and transmits when found idle.

P-persistent: The node senses the medium, if idle it sends the data with p probability. If the data is not transmitted ((1-p) probability) then it waits for some time and checks the medium again, now if it is found idle then it send with p probability. This repeat continues until the frame is sent. It is used in Wifi and packet radio systems.

O-persistent: Superiority of nodes is decided beforehand and transmission occurs in that order. If the medium is idle, node waits for its time slot to send data.

(c) CSMA/CD – Carrier sense multiple access with collision detection. Stations can terminate transmission of data if collision is detected.

(d) CSMA/CA – Carrier sense multiple access with collision avoidance. The process of collisions detection involves sender receiving acknowledgement signals. If there is just one signal(its own) then the data is successfully sent but if there are two signals(its own and the one with which it has collided) then it means a collision has occurred. To distinguish between these two cases, collision must have a lot of impact on received signal. However it is not so in wired networks, so CSMA/CA is used in this case.

2. Controlled Access:
In this, the data is sent by that station which is approved by all other stations.

3. Channelization:
In this, the available bandwidth of the link is shared in time, frequency and code to multiple stations to access channel simultaneously.

Frequency Division Multiple Access (FDMA) – The available bandwidth is divided into equal bands so that each station can be allocated its own band. Guard bands are also added so that no two bands overlap to avoid crosstalk and noise.

Time Division Multiple Access (TDMA) – In this, the bandwidth is shared between multiple stations. To avoid collision time is divided into slots and stations are allotted these slots to transmit data. However there is a overhead of synchronization as each station needs to know its time slot. This is resolved by adding synchronization bits to each slot. Another issue with TDMA is propagation delay which is resolved by addition of guard bands.

Code Division Multiple Access (CDMA) – One channel carries all transmissions simultaneously. There is neither division of bandwidth nor division of time. For example, if there are many people in a room all speaking at the same time, then also perfect reception of data is possible if only two person speak the same language. Similarly, data from different stations can be transmitted simultaneously in different code languages.

Orthogonal Frequency Division Multiple Access (OFDMA) – In OFDMA the available bandwidth is divided into small subcarriers in order to increase the overall performance, Now the data is transmitted through these small subcarriers. it is widely used in the 5G technology.

- **Bridges**

Bridged Ethernet, also known as Ethernet bridging or transparent bridging, is a networking technique used to connect multiple LAN segments together into a single logical network. It involves the use of network bridges, which operate at the data link layer (Layer 2) of the OSI model.

Here's how bridged Ethernet works:

1. Network Bridges: A network bridge is a networking device that connects two or more LAN segments, allowing them to communicate with each other as if they were part of the same LAN. Bridges work by examining the MAC addresses of incoming Ethernet frames and making forwarding decisions based on the destination MAC address.
2. MAC Learning: When a bridge receives an Ethernet frame, it examines the source MAC address and associates it with the incoming port. This process is called MAC learning. The bridge maintains a table, known as the MAC address table or forwarding table, which maps MAC addresses to specific bridge ports.
3. Forwarding Decisions: When a bridge receives an Ethernet frame with a destination MAC address, it looks up the MAC address table to determine the outgoing port for the frame. If the destination MAC address is not found in the table, the bridge forwards the frame to all ports except the incoming port (flooding), allowing the frame to reach its destination.
4. Filtering and Forwarding: Once a bridge has learned the MAC addresses of devices on each LAN segment, it can make filtering decisions to optimize network traffic. When a bridge receives an Ethernet frame, it checks the destination MAC address against the MAC address table. If the destination MAC address is known, the bridge forwards the frame only to the port associated with that MAC address. This process is called unicast forwarding.
5. Loop Prevention: Bridged Ethernet networks must prevent loops, which can lead to broadcast storms and network congestion. To avoid loops, bridges use the Spanning Tree Protocol (STP) or Rapid Spanning Tree Protocol (RSTP). These protocols establish a loop-free topology by selectively blocking redundant paths while maintaining alternate paths for redundancy.

By connecting LAN segments with bridges, bridged Ethernet networks provide several benefits:

- Increased network capacity: Bridges enable the expansion of a network by connecting multiple LAN segments, allowing for more devices and increased bandwidth.
- Traffic isolation: Each LAN segment connected by a bridge forms a separate collision domain, isolating traffic and reducing network congestion.
- MAC address learning and filtering: Bridges learn and store MAC addresses, enabling efficient forwarding of frames only to the intended destination.
- Loop prevention: Spanning Tree Protocol prevents loops in bridged networks, ensuring stable and reliable network operation.

Bridged Ethernet is commonly implemented using network switches, which integrate bridge functionality into their design. Switches offer multiple ports to connect devices and perform MAC address learning and forwarding, providing a seamless and efficient bridged Ethernet network.

- **Switched Ethernet**

Switched Ethernet, also known as Ethernet switching or switched LAN, is a networking technology that uses network switches to connect devices in a local area network (LAN). It provides a fast and efficient way of forwarding Ethernet frames between devices within a LAN. Switched Ethernet operates at the data link layer (Layer 2) of the OSI model.

Here's how switched Ethernet works:

1. Network Switches: A network switch is a networking device that receives Ethernet frames from connected devices and forwards them to their intended destinations. Switches have multiple ports, typically ranging from a few ports to dozens or more, allowing multiple devices to be connected simultaneously.
2. MAC Address Learning: When an Ethernet frame arrives at a switch, it examines the source MAC address in the frame and associates it with the incoming port. This process is called MAC address learning. The switch maintains a table, known as the MAC address table or forwarding table, which maps MAC addresses to specific switch ports.
3. Frame Forwarding: When a switch receives an Ethernet frame, it looks up the destination MAC address in the MAC address table. If the destination MAC address is found in the table, the switch forwards the frame only to the port associated with that MAC address. This process is known as unicast forwarding.

If the destination MAC address is not found in the table, the switch forwards the frame to all ports except the incoming port (flooding), allowing the frame to reach its destination. This is similar to the behavior of bridges in bridged Ethernet networks.

4. Switching Efficiency: Switched Ethernet provides efficient and high-performance data forwarding. Unlike hubs, which simply broadcast incoming frames to all connected devices, switches selectively forward frames based on MAC address information. This significantly reduces network congestion and improves overall network performance.
5. Virtual LANs (VLANs): Switched Ethernet supports the concept of virtual LANs (VLANs), which allow the logical segmentation of a physical LAN into multiple virtual LANs. VLANs enable the isolation of network traffic and provide enhanced security and flexibility. Devices within the same VLAN can communicate with each other as if they were on the same physical LAN, regardless of their physical location within the network.
6. Spanning Tree Protocol (STP): To prevent loops and ensure network stability in switched Ethernet networks, the Spanning Tree Protocol (STP) or its faster variant, Rapid Spanning Tree Protocol (RSTP), is used. These protocols create a loop-free topology by selectively blocking redundant paths while keeping alternate paths for redundancy. This prevents broadcast storms and network congestion caused by loops.

Switched Ethernet has become the dominant technology for LAN connectivity due to its high performance, scalability, and flexibility. It allows devices to communicate efficiently within a LAN by selectively forwarding Ethernet frames based on MAC addresses, providing a reliable and fast data transfer within the network.

- **Full Duplex Ethernet**

Full Duplex Ethernet is a communication mode in Ethernet networks that allows simultaneous bidirectional data transfer between two devices without the need for collisions. It provides independent data transmission paths in both directions, allowing devices to send and receive data at the same time. Full Duplex Ethernet operates at the physical layer (Layer 1) and data link layer (Layer 2) of the OSI model.

Here's how Full Duplex Ethernet works:

1. Traditional Ethernet: In traditional Ethernet, also known as half-duplex Ethernet, devices share the same communication medium (e.g., twisted pair, coaxial cable) and use the Carrier Sense Multiple Access with Collision Detection (CSMA/CD) protocol. In this mode, devices take turns transmitting data and listen for collisions. If a collision occurs, devices involved in the collision wait for a random period before retransmitting.
2. Collision Avoidance: Full Duplex Ethernet eliminates the need for collision detection and avoids collisions altogether. With full duplex, devices have separate dedicated communication paths for transmitting and receiving data, removing the possibility of collisions.
3. Simultaneous Transmission and Reception: In Full Duplex Ethernet, devices can transmit and receive data simultaneously, allowing for maximum utilization of the available bandwidth. This bidirectional communication mode enables faster and more efficient data transfer between devices.
4. Dedicated Communication Paths: To achieve full duplex communication, devices need to be connected through a medium that supports separate transmit and receive channels. This can be accomplished using two pairs of wires in twisted pair Ethernet or separate fibers in fiber optic Ethernet. Network switches and network interface cards (NICs) must also support full duplex operation.
5. Duplex Negotiation: When devices are connected, they perform a duplex negotiation process to determine whether full duplex operation is supported. The devices exchange signaling information called link pulses to establish full duplex capability. If both devices support full duplex, they can operate in full duplex mode. Otherwise, they fall back to half-duplex operation.

Benefits of Full Duplex Ethernet:

- Increased Throughput: Full duplex allows for simultaneous transmission and reception, effectively doubling the potential throughput compared to half-duplex Ethernet.
- Lower Latency: With full duplex, devices can transmit and receive data without waiting for turnarounds or collision detection, resulting in lower latency and faster communication.
- Improved Efficiency: Full duplex eliminates collisions, reducing retransmissions and increasing network efficiency. This is particularly beneficial in high-demand environments where multiple devices need to communicate simultaneously.

- Enhanced Performance: Full duplex enables continuous, uninterrupted communication between devices, supporting applications that require real-time data transfer or low-latency communication.

Full Duplex Ethernet has become the standard in modern Ethernet networks, enabling faster and more efficient data transmission by providing separate transmit and receive channels. It offers significant advantages over traditional half-duplex Ethernet, making it suitable for a wide range of applications, from home networks to enterprise environments.

- **Fast & gigabit Ethernet**

Fast Ethernet and Gigabit Ethernet are two different standards for Ethernet networks that provide higher data transmission speeds compared to traditional Ethernet. Let's explore each of them:

1. Fast Ethernet (IEEE 802.3u): Fast Ethernet extends the original Ethernet standard to provide higher data transmission rates. It operates at 100 Mbps (megabits per second), which is ten times faster than traditional Ethernet's 10 Mbps speed. Fast Ethernet uses the same CSMA/CD (Carrier Sense Multiple Access with Collision Detection) protocol as Ethernet for medium access control.

Fast Ethernet can be implemented using different physical media, including twisted pair copper cables (e.g., Cat 5 or Cat 5e) and fiber optic cables. It offers backward compatibility with traditional Ethernet, allowing Fast Ethernet devices to communicate with Ethernet devices. Fast Ethernet switches are widely available and commonly used for local area networks (LANs).

2. Gigabit Ethernet (IEEE 802.3ab): Gigabit Ethernet is a further advancement in Ethernet technology, providing data transmission rates of 1 Gbps (gigabit per second). It is ten times faster than Fast Ethernet and a hundred times faster than traditional Ethernet. Gigabit Ethernet operates in full-duplex mode, allowing simultaneous bidirectional communication.

Gigabit Ethernet supports various physical media, including twisted pair copper cables (e.g., Cat 5e or Cat 6) and fiber optic cables. It uses the same CSMA/CD protocol as Ethernet and Fast Ethernet, although collision detection is not utilized in full-duplex mode.

Gigabit Ethernet is widely deployed in LANs and has become the standard for many network applications due to its high data transmission rates. It provides faster and more efficient network performance, especially for bandwidth-intensive tasks such as large file transfers, multimedia streaming, and high-speed data processing.

It's worth noting that both Fast Ethernet and Gigabit Ethernet are backward compatible with Ethernet. This means that devices supporting higher-speed Ethernet standards can still communicate with devices using lower-speed Ethernet standards.

Overall, Fast Ethernet and Gigabit Ethernet are advancements in Ethernet technology, offering significantly higher data transmission rates compared to traditional Ethernet. They have played a crucial role in meeting the increasing demands for faster and more reliable network connectivity in various environments.

- **Introduction to Data link layer in 802.11 LAN**

The Data Link Layer is the second layer of the OSI (Open Systems Interconnection) model, responsible for the reliable transfer of data between nodes over a physical medium. In the context of 802.11 LANs (wireless local area networks), the Data Link Layer plays a crucial role in ensuring the reliable transmission of data frames between wireless devices.

The Data Link Layer in 802.11 LANs is often divided into two sublayers: the Logical Link Control (LLC) sublayer and the Media Access Control (MAC) sublayer. Let's explore each of these sublayers:

1. Logical Link Control (LLC) Sublayer: The LLC sublayer is responsible for managing data link connections between devices in an 802.11 LAN. It provides a reliable and error-free communication channel by implementing protocols for flow control, error control, and addressing. The LLC sublayer is independent of the physical medium and focuses on the logical aspects of data transfer.

One common LLC protocol used in 802.11 LANs is the IEEE 802.2 standard, which defines the format of data frames, addressing schemes, and mechanisms for error detection and correction.

2. Media Access Control (MAC) Sublayer: The MAC sublayer is responsible for controlling access to the wireless medium and handling data transmission between devices within an 802.11 LAN. It ensures that multiple devices can share the limited bandwidth of the wireless channel efficiently and fairly.

The MAC sublayer in 802.11 LANs uses the Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) protocol. Unlike wired Ethernet networks that use collision detection (CSMA/CD), CSMA/CA is designed to avoid collisions in wireless environments.

Network Devices: Network devices, also known as networking hardware, are physical devices that allow hardware on a computer network to communicate and interact with one another. For example Repeater, Hub, Bridge, Switch, Routers, Gateway, Brouter, and NIC, etc.

1. Repeater – A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted to extend the length to which the signal can be transmitted over the same network. An important point to be noted about repeaters is that they not only amplify the signal but also regenerate it. When the signal becomes weak, they copy it bit by bit and regenerate it at its star topology connectors connecting following the original strength. It is a 2-port device.

2. Hub – A hub is a basically multi-port repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, the collision domain of all hosts connected through Hub remains one. Also, they do not have the intelligence to find out the best path for data packets which leads to inefficiencies and wastage.

Types of Hub

Active Hub:- These are the hubs that have their power supply and can clean, boost, and relay the signal along with the network. It serves both as a repeater as well as a wiring center. These are used to extend the maximum distance between nodes.

Passive Hub:- These are the hubs that collect wiring from nodes and power supply from the active hub. These hubs relay signals onto the network without cleaning and boosting them and can’t be used to extend the distance between nodes.

Intelligent Hub:- It works like an active hub and includes remote management capabilities. They also provide flexible data rates to network devices. It also enables an administrator to monitor the traffic passing through the hub and to configure each port in the hub.

3. Bridge – A bridge operates at the data link layer. A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of the source and destination. It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a 2 port device.

Types of Bridges

Transparent Bridges:- These are the bridge in which the stations are completely unaware of the bridge's existence i.e. whether or not a bridge is added or deleted from the network, reconfiguration of the stations is unnecessary. These bridges make use of two processes i.e. bridge forwarding and bridge learning.

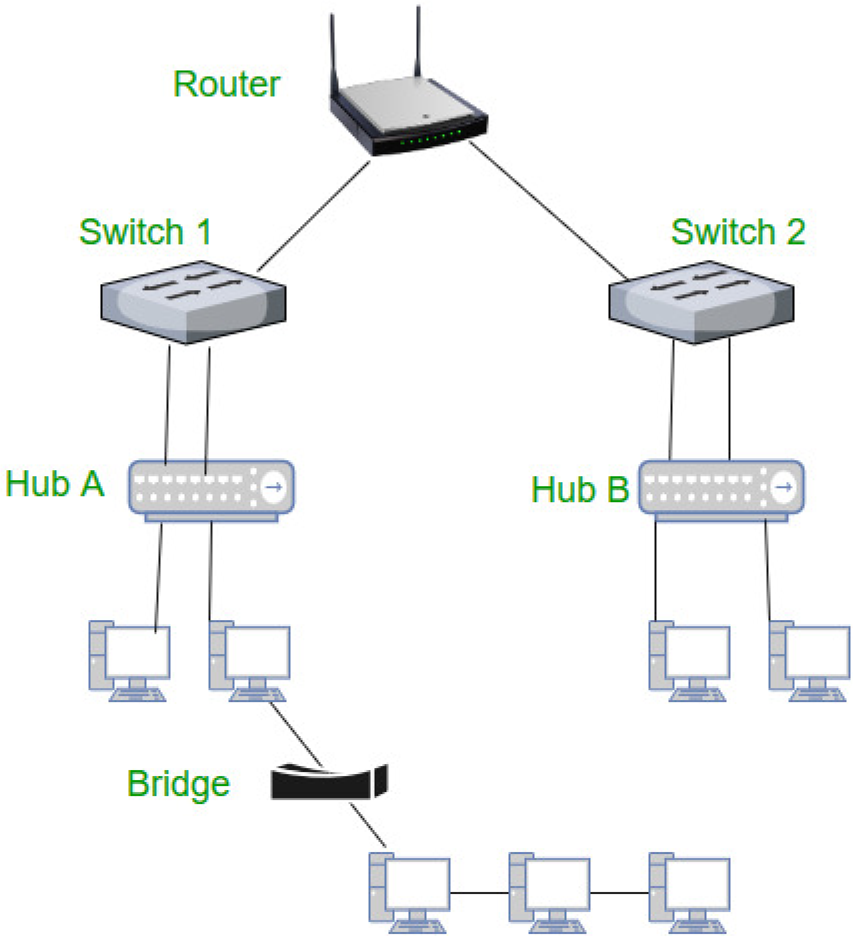
Source Routing Bridges:- In these bridges, routing operation is performed by the source station and the frame specifies which route to follow. The host can discover the frame by sending a special frame called the discovery frame, which spreads through the entire network using all possible paths to the destination.

4. Switch – A switch is a multiport bridge with a buffer and a design that can boost its efficiency(a large number of ports imply less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, which makes it very efficient as it does not forward packets that have errors and forward good packets selectively to the correct port only. In other words, the switch divides the collision domain of hosts, but the broadcast domain remains the same.

Types of Switch

- Unmanaged switches: These switches have a simple plug-and-play design and do not offer advanced configuration options. They are suitable for small networks or for use as an expansion to a larger network.
- Managed switches: These switches offer advanced configuration options such as VLANs, QoS, and link aggregation. They are suitable for larger, more complex networks and allow for centralized management.
- Smart switches: These switches have features similar to managed switches but are typically easier to set up and manage. They are suitable for small- to medium-sized networks.
- Layer 2 switches: These switches operate at the Data Link layer of the OSI model and are responsible for forwarding data between devices on the same network segment.
- Layer 3 switches: These switches operate at the Network layer of the OSI model and can route data between different network segments. They are more advanced than Layer 2 switches and are often used in larger, more complex networks.
- PoE switches: These switches have Power over Ethernet capabilities, which allows them to supply power to network devices over the same cable that carries data.
- Gigabit switches: These switches support Gigabit Ethernet speeds, which are faster than traditional Ethernet speeds.
- Rack-mounted switches: These switches are designed to be mounted in a server rack and are suitable for use in data centers or other large networks.
- Desktop switches: These switches are designed for use on a desktop or in a small office environment and are typically smaller in size than rack-mounted switches.
- Modular switches: These switches have modular design, which allows for easy expansion or customization. They are suitable for large networks and data centers.

5. Routers – A router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANs and have a dynamically updating routing table based on which they make decisions on routing the data packets. The router divides the broadcast domains of hosts connected through it.



6. Gateway – A gateway, as the name suggests, is a passage to connect two networks that may work upon different networking models. They work as messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switches or routers. A gateway is also called a protocol converter.

- **Backbone**

Backbone is most important part of a system which provides the central support to the rest system, for example backbone of a human body that balance and hold all the body parts. Similarly in Computer Networks a Backbone Network is as a Network containing a high capacity connectivity infrastructure that backbone to the different part of the network.

Actually a backbone network allows multiple LANs to get connected in a backbone network, not a single station is directly connected to the backbone but the stations are part of LAN, and backbone connect those LANs.

Backbone LANs:
Because of increasing use of distributed applications and PCs, a new flexible strategy for LANs has been introduced. if a premises wide data communication system is to be supported then we need a networking system which can span over the required distance and which capable of interconnecting all the equipment in a single building or in a group of buildings.

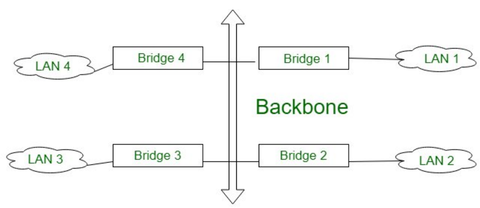
It is possible to develop a single LAN for this purpose but practically this scheme faces the following drawbacks:

- Poor Reliability:
  With a single LAN, the reliability will be poor since a service interruption even for a short duration can cause major problem to the user.
- Capacity:
  There is a possibility that a single LAN may be saturated due to increase in number of devices beyond a certain number
- Cost:
  A single LAN can not give its optimum performance for the diverse requirements of communication and interconnection.

So the alternative for using a single LAN is to use low cost low capacity LANs in each building or department and then interconnection all these LANs with high capacity LAN. such a network is called as Backbone LAN. the backbone network allows several LANs to be connected. in the backbone network, no station is directly connected with backbone, instead each station is a part of a LAN, and the LANs are connected to the backbone.

The backbone itself is a LAN, it uses a LAN protocol such as ethernet, Hence each connection in the backbone is itself another LAN. The two very common used architectures are: Bus backbone, Star backbone. These are explained as following below.

Bus Backbone:
In Bus backbone the topology used for the backbone is bus topology.

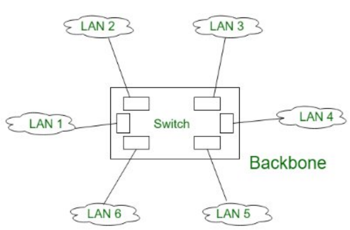


Structure of a Bus
backbone

In above the Bus backbone structure is used as a distribution backbone for connecting different buildings in an organization. each building may have either a single LAN or another backbone which comes in star backbone. the structure is a bridge based (bridge is the connecting device) backbone with four LANs.

Working:
In above structure if a station in LAN 2 wants to send a frame to some other station in Same LAN then Bridge 2 will not allow the frame to pass to any other LAN, hence this frame will not reach the backbone. If a station from LAN 1 wants to send a frame to a station in LAN 4 then Bridge 1 passes this frame to the backbone. This frame is then received by Bridge 4 and delivered to the destination.

Star Backbone:
The topology of this backbone is star topology.

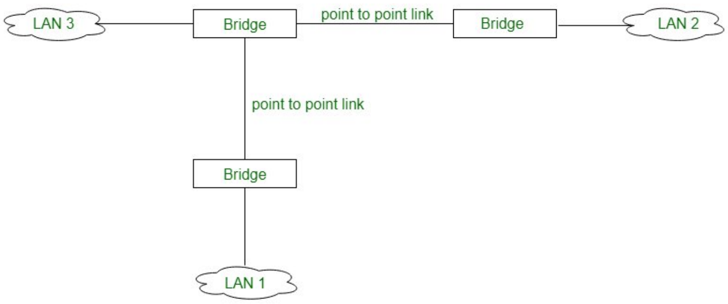


Star Backbone

Above figure shows the Star backbone in this configuration, the backbone is simply a switch which is used to connect various LANs. The switch does the job of backbone and connect the LANs as well. This type of backbone are basically used as distribution backbone inside a building.

There is one more category of backbone network is Interconnecting of Remote LANs:

Interconnection of Remote control:
In this type of backbone network the connection are done through the bridge called remote bridges which acts as connecting devices in connect LANs as point to point network link.



Connecting remote LANs to each other

Example of point to point networks are leased telephone lines or ADLS lines. Such a point to point network can be considered as being equivalent ta a LAN without stations.

- **VLAN**

Virtual LAN (VLAN) is a concept in which we can divide the devices logically on layer 2 (data link layer). Generally, layer 3 devices divide the broadcast domain but the broadcast domain can be divided by switches using the concept of VLAN.

A broadcast domain is a network segment in which if a device broadcast a packet then all the devices in the same broadcast domain will receive it. The devices in the same broadcast domain will receive all the broadcast packets but it is limited to switches only as routers don't forward out the broadcast packet. To forward out the packets to different VLAN (from one VLAN to another) or broadcast domains, inter Vlan routing is needed. Through VLAN, different small-size sub-networks are created which are comparatively easy to handle.

VLAN ranges:

- VLAN 0, 4095: These are reserved VLAN which cannot be seen or used.
- VLAN 1: It is the default VLAN of switches. By default, all switch ports are in VLAN. This VLAN can't be deleted or edit but can be used.
- VLAN 2-1001: This is a normal VLAN range. We can create, edit and delete these VLAN.
- VLAN 1002-1005: These are CISCO defaults for fddi and token rings. These VLAN can't be deleted.
- Vlan 1006-4094: This is the extended range of Vlan.

VLANs offer several features and benefits, including:

- Improved network security: VLANs can be used to separate network traffic and limit access to specific network resources. This improves security by preventing unauthorized access to sensitive data and network resources.
- Better network performance: By segregating network traffic into smaller logical networks, VLANs can reduce the amount of broadcast traffic and improve network performance.
- Simplified network management: VLANs allow network administrators to group devices together logically, rather than physically, which can simplify network management tasks such as configuration, troubleshooting, and maintenance.

- Flexibility: VLANs can be configured dynamically, allowing network administrators to quickly and easily adjust network configurations as needed.
- Cost savings: VLANs can help reduce hardware costs by allowing multiple virtual networks to share a single physical network infrastructure.
- Scalability: VLANs can be used to segment a network into smaller, more manageable groups as the network grows in size and complexity.